Homework 9 - Charlie Caravati

In [2]:
```
import matplotlib.pyplot as plt
import numpy as np
import numpy.linalg as la
import math
from math import factorial
from numpy.linalg import inv
from numpy.linalg import norm
```

1.

$$\begin{bmatrix}1 & 0\\0 & 1\\0 & 1\end{bmatrix}\begin{bmatrix}u\\v\end{bmatrix}=\begin{bmatrix}1\\1\\0\end{bmatrix}$$

For $A=\begin{bmatrix}1 & 0\\0 & 1\\0 & 1\end{bmatrix}, A^T=\begin{bmatrix}1 & 0 & 0\\0 & 1 & 1\end{bmatrix}$

$$A^TA=\begin{bmatrix}1 & 0\\0 & 1\\0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\0 & 1 & 1\end{bmatrix}=\begin{bmatrix}1 & 0\\0 & 2\end{bmatrix}$$

For $b=\begin{bmatrix}1\\1\\0\end{bmatrix}$

$$A^Tb=\begin{bmatrix}1 & 0 & 0\\0 & 1 & 1\end{bmatrix}\begin{bmatrix}1\\1\\0\end{bmatrix}=\begin{bmatrix}1\\1\end{bmatrix}$$

$$\begin{bmatrix}1 & 0\\0 & 2\end{bmatrix}\begin{bmatrix}u\\v\end{bmatrix}=\begin{bmatrix}1\\1\end{bmatrix}$$

$1(u)+0(v)=1\rightarrow u=1$

$0(u)+2(v)=1\rightarrow v=1/2$

$$x=\begin{bmatrix}u\\v\end{bmatrix}\begin{bmatrix}1\\1/2\end{bmatrix}$$

2.

Minimize $E^2=b_1^2+4b_2^2+25b_3^2+9b_4^2$ when

$$\begin{bmatrix}1 & 3\\6 & -1\\4 & 0\\2 & 7\end{bmatrix}\begin{bmatrix}x_1\\x_2\end{bmatrix}-\begin{bmatrix}1\\2\\3\\4\end{bmatrix}=\begin{bmatrix}b_1\\b_2\\b_3\\b_4\end{bmatrix}$$

$Ax-c=b$ minimizing by setting $b=0$, rearranging $Ax=c$, and multiplying both sides by the diagonal of the square root of the coefficients of the error function in order to use the normal functions to solve $(DA)^T(DA)x=(DA)^T(Dc)$

$$D=\begin{bmatrix}1 & 0 & 0 & 0\\0 & 2 & 0 & 0\\0 & 0 & 5 & 0\\0 & 0 & 0 & 3\end{bmatrix}$$

$$DA=\begin{bmatrix}1 & 0 & 0 & 0\\0 & 2 & 0 & 0\\0 & 0 & 5 & 0\\0 & 0 & 0 & 3\end{bmatrix}\begin{bmatrix}1 & 3\\6 & -1\\4 & 0\\2 & 7\end{bmatrix}=\begin{bmatrix}1(1) & 3(1)\\6(2) & -1(2)\\4(5) & 0(5)\\2(3) & 7(3)\end{bmatrix}=\begin{bmatrix}1 & 3\\12 & -2\\20 & 0\\6 & 21\end{bmatrix}$$

$$(DA)^T=\begin{bmatrix}1 & 12 & 20 & 6\\3 & -2 & 0 & 21\end{bmatrix}$$

$$Dc=\begin{bmatrix}1 & 0 & 0 & 0\\0 & 2 & 0 & 0\\0 & 0 & 5 & 0\\0 & 0 & 0 & 3\end{bmatrix}\begin{bmatrix}1\\2\\3\\4\end{bmatrix}=\begin{bmatrix}1\\4\\15\\12\end{bmatrix}$$

$(DA)^T(DA)x=(DA)^TDc$

$$(DA)^T(DA)=\begin{bmatrix}1 & 12 & 20 & 6\\3 & -2 & 0 & 21\end{bmatrix}\begin{bmatrix}1 & 3\\12 & -2\\20 & 0\\6 & 21\end{bmatrix}=\begin{bmatrix}1^2+12^2+20^2+6^2 & 1(3)+12(-2)+20(0)+6(21)\\1(3)+12(-2)+20(0)+6(21) & 3^2+(-2)^2+0^2+21^2\end{bmatrix}=\begin{bmatrix}581 & 105\\105 & 454\end{bmatrix}$$

$$(DA)^T(Dc)=\begin{bmatrix}1 & 12 & 20 & 6\\3 & -2 & 0 & 21\end{bmatrix}\begin{bmatrix}1\\4\\15\\12\end{bmatrix}=\begin{bmatrix}1(1)+12(4)+20(15)+6(12)\\3(1)-2(4)+0(15)+21(12)\end{bmatrix}=\begin{bmatrix}421\\247\end{bmatrix}$$

$$\begin{bmatrix} 581 & 105 \\ 105 & 454 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 421 \\ 247 \end{bmatrix}$$

Using a matrix solver, $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.65361 \\ 0.39289 \end{bmatrix}$

3.

Prove three term recursion formula for orthogonal polynomials

$\phi_k(x) = (x - b_k)\phi_{k-1}(x) - c_k\phi_{k-2}(x)$

where $b_k = \frac{<x\phi_{k-1}, \phi_{k-1}>}{<\phi_{k-1}, \phi_{k-1}>}, c_k = \frac{<x\phi_{k-1}, \phi_{k-2}>}{<\phi_{k-2}, \phi_{k-2}>}$

(i) $\phi_k(x) = (x - b_k)\phi_{k-1}(x) - c_k\phi_{k-2}(x) - \{a_{k-3}\phi_{k-3}(x) + a_{k-4}\phi_{k-4}(x) + \cdots + a_0\phi_0(x)\}$

$\phi_k(x) = x\phi_{k-1}(x) - b_k\phi_{k-1}(x) - c_k\phi_{k-2}(x) - \sum_{j=0}^{k-3} a_j\phi_j(x)$

Taking the inner product with a specific $\phi_j(x)$ for $j \leq k - 3$

$< \phi_k, \phi_j >=< x\phi_{k-1}, \phi_j > -b_k < \phi_{k-1}, \phi_j > -c_k < \phi_{k-2}, \phi_j > -a_j < \phi_j, \phi_j >$

$< \phi_k, \phi_j >= 0$ since $j < k$ and the definitely of orthogonal

$< x\phi_{k-1}, \phi_j >=< \phi_{k-1}, x\phi_j >$ where $x\phi_j$ has order $j + 1 < k - 1$ then $< x\phi_{k-1}, \phi_j >= 0$

$< \phi_{k-1}, \phi_j >= 0$ and $< \phi_{k-2}, \phi_j >= 0$ as well since $j < k - 2 < k - 1$

$\rightarrow 0 = 0 - b_k(0) - c_k(0) - a_j < \phi_j, \phi_j > \rightarrow a_j < \phi_j, \phi_j >= 0$ for all a_j which forces $a_j = 0$ for all $j \leq k - 3$

Which gives $\phi_k(x) = x\phi_{k-1}(x) - b_k\phi_{k-1}(x) - c_k\phi_{k-2}(x)$

Taking the inner product with $\phi_{k-1}(x)$ gives

$< \phi_k, \phi_{k-1} >=< x\phi_{k-1}, \phi_{k-1} > -b_k < \phi_{k-1}, \phi_{k-1} > -c_k < \phi_{k-2}, \phi_{k-1} >$

$< \phi_k, \phi_{k-1} >= 0$ and $< \phi_{k-2}, \phi_{k-1} >= 0$ gives

$0 =< x\phi_{k-1}, \phi_{k-1} > -b_k < \phi_{k-1}, \phi_{k-1} >$ gives $< x\phi_{k-1}, \phi_{k-1} >= b_k < \phi_{k-1}, \phi_{k-1} >$

$b_k = \frac{<x\phi_{k-1}, \phi_{k-1}>}{<\phi_{k-1}, \phi_{k-1}>}$

Taking the inner product with $\phi_{k-2}(x)$ gives

$< \phi_k, \phi_{k-2} >=< x\phi_{k-1}, \phi_{k-2} > -b_k < \phi_{k-1}, \phi_{k-2} > -c_k < \phi_{k-2}, \phi_{k-2} >$

Where $< \phi_k, \phi_{k-2} >= 0, < \phi_{k-1}, \phi_{k-2} >= 0$ gives

$0 =< x\phi_{k-1}, \phi_{k-2} > -c_k < \phi_{k-2}, \phi_{k-2} >$ gives $< x\phi_{k-1}, \phi_{k-2} >= c_k < \phi_{k-2}, \phi_{k-2} >$ gives

$c_k = \frac{<x\phi_{k-1}, \phi_{k-2}>}{<\phi_{k-2}, \phi_{k-2}>}$

4.

Padé Approximation of $f(x) = sin(x)$

(a) M = N = 3

$P_3^3 = \frac{a_0 + a_1x + a_2x^2 + a_3x^3}{1 + b_1x + b_2x^2 + b_3x^3}$

$T_6 = x - \frac{1}{6}x^3 + \frac{1}{120}x^5$

$a_0 + a_1x + a_2x^2 + a_3x^3 = (x - \frac{1}{6}x^3 + \frac{1}{120}x^5)(1 + b_1x + b_2x^2 + b_3x^3)$

$x^0 : 0$

$x^1 : 1x$

$x^2 : b_1x^2$

$x^3 : (b_2 - \frac{1}{6})x^3$

$x^4 : (b_3 - \frac{b_1}{6})x^4$

$x^5 : (\frac{1}{120} - \frac{b_2}{6})x^5$

$x^6 : (\frac{b_1}{120} - \frac{b_3}{6})x^6$

$a_0 = 0$

$a_1 = 1$

$a_2 = b_1$

$a_3 = (b_2 - \frac{1}{6}) = \frac{1}{20} - \frac{1}{6} = -\frac{7}{60}$

$(b_3 - \frac{b_1}{6}) = 0$

$(\frac{1}{120} - \frac{b_2}{6}) = 0$

$(\frac{b_1}{120} - \frac{b_3}{6}) = 0$

$b_2 = \frac{6}{120} = \frac{1}{20}$

$b_3 = \frac{b_1}{6} \rightarrow (\frac{b_1}{120} - \frac{b_1/6}{6}) = (\frac{b_1}{120} - \frac{b_1}{36}) = 0$

$b_1(\frac{1}{120} - \frac{1}{36}) = 0 \rightarrow b_1 = 0$

$(b_3 - \frac{0}{6}) = 0 \rightarrow b_3 = 0$

$P_3^3 = \frac{0+1x+(0)x^2+-\frac{7}{60}x^3}{1+0x+\frac{1}{20}x^2+0x^3} = \frac{x-\frac{7}{60}x^3}{1+\frac{1}{20}x^2}$

> (b) M = 4, N = 2

$P_2^4 = \frac{a_0+a_1x+a_2x^2+a_3x^3+a_4x^4}{1+b_1x+b_2x^2}$

$T_6 = x - \frac{1}{6}x^3 + \frac{1}{120}x^5$

$a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 = (x - \frac{1}{6}x^3 + \frac{1}{120}x^5)(1 + b_1x + b_2x^2)$

$x^0 : 0$

$x^1 : 1x$

$x^2 : b_1x^2$

$x^3 : (b_2 - \frac{1}{6})x^3$

$x^4 : (-\frac{b_1}{6})x^4$

$x^5 : (\frac{1}{120} - \frac{b_2}{6})x^5$

$x^6 : (\frac{b_1}{120})x^6$

$a_0 = 0$

$a_1 = 1$

$a_2 = b_1 = 0$

$a_3 = (b_2 - \frac{1}{6}) = \frac{1}{20}\frac{1}{6} = -\frac{7}{60}$

$a_4 = (-\frac{b_1}{6}) = 0/6 = 0$

$(\frac{1}{120} - \frac{b_2}{6}) = 0$

$(\frac{b_1}{120}) = 0$

$b_2 = \frac{6}{120} = \frac{1}{20}$

$b_1(\frac{1}{120}) = 0 \rightarrow b_1 = 0$

$P_4^2 = \frac{0+1x+(0)x^2+-\frac{7}{60}x^3+(0)x^4}{1+0x+\frac{1}{20}x^2} = \frac{x-\frac{7}{60}x^3}{1+\frac{1}{20}x^2}$ which matches P_3^3 in (a)

> (c) M = 2, N = 4

$P_4^2 = \frac{a_0+a_1x+a_2x^2}{1+b_1x+b_2x^2+b_3x^3+b_4x^4}$

$T_6 = x - \frac{1}{6}x^3 + \frac{1}{120}x^5$

$a_0 + a_1x = (x - \frac{1}{6}x^3 + \frac{1}{120}x^5)(1 + b_1x + b_2x^2 + b_3x^3 + b_4x^4)$

$x^0 : 0$

$x^1 : 1x$

$x^2 : b_1x^2$

$x^3 : (b_2 - \frac{1}{6})x^3$

$x^4 : (b_3 - \frac{b_1}{6})x^4$

$x^5 : (b_4 + \frac{1}{120} - \frac{b_2}{6})x^5$

$x^6 : (\frac{b_1}{120} - \frac{b_3}{6})x^6$

$a_0 = 0$

$a_1 = 1$

$a_2 = b_1 = 0$

$b_2 - \frac{1}{6} = 0$

$b_3 - \frac{b_1}{6} = 0$

$b_4 + \frac{1}{120} - \frac{b_2}{6} = 0$

$\frac{b_1}{120} - \frac{b_3}{6} = 0$

$b_2 = \frac{1}{6}$

$b_3 = \frac{b_1}{6} \to \frac{b_1}{120} - \frac{b_1/6}{6} = \frac{b_1}{120} - \frac{b_1}{36} \to b_1(1/120 - 1/36) = 0 \to b_1 = 0$

$b_3 - \frac{b_1}{6} = 0 \to b_3 - 0 = 0 \to b_3 = 0$

$b_4 + \frac{1}{120} - \frac{1/6}{6} = 0 \to b_4 = -\frac{1}{120} + \frac{1}{36} = \frac{7}{360}$

$P_4^2 = \frac{0+1x+(0)x^2}{1+0x+\frac{1}{6}x^2+0x^3+\frac{7}{360}x^4} = \frac{x}{1+\frac{1}{6}x^2+\frac{7}{360}x^4}$

In [6]:
```
x = np.linspace(0, 5, 500)

#true
true_y = np.sin(x)

# 6th order maclaurin
m6 = x - (x ** 3) / 6 + (x ** 5) / 120

# (a) & (b)
pade_33 = (x - (7 / 60) * (x ** 3)) / (1 + (1 / 20) * (x ** 2))

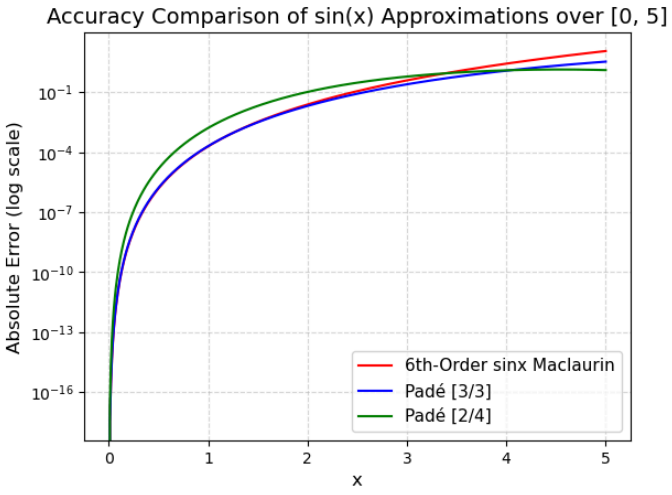
# (c)
pade_24 = x / (1 + (1 / 6) * (x ** 2) + (7 / 360) * (x ** 4))

# calc errors
error_mac = np.abs(true_y - m6)
error_pade_33 = np.abs(true_y - pade_33)
error_pade_24 = np.abs(true_y - pade_24)

# plot
plt.plot(x, error_mac, label="6th-Order sinx Maclaurin", color="red")
plt.plot(x, error_pade_33, label="Padé [3/3]", color="blue")
plt.plot(x, error_pade_24, label="Padé [2/4]", color="green")

plt.yscale("log")
plt.xlabel("x", fontsize=12)
plt.ylabel("Absolute Error (log scale)", fontsize=12)
plt.title(
    "Accuracy Comparison of sin(x) Approximations over [0, 5]", fontsize=14
)
plt.legend(fontsize=11)
plt.grid(True, which="both", ls="--", alpha=0.5)

plt.show()
```



The error for the approximation and Maclaurin expansion are very similar over the interval, which maxes sense as all three errors are bounded by $O(x^7)$. The P_3^3 approximation seems to start showing a clearer advantage near the end of the interval but the gap is still small.